Mingyu Ouyang

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Research Focus Areas

My research focuses on multimodal digital agents that can perceive, plan, and ground their actions in digital environments. I am particularly interested in GUI, computer-use, and game agents, as well as the fine-grained perception and grounding required for reliable agent-environment interaction.

Education

National University of Singapore — **Singapore**
Ph.D. in Computer Science — *Jan. 2025 – Present*
Supervisors: Prof. Hwee Tou Ng and Prof. Mike Zheng Shou

National University of Singapore — **Singapore**
M.Sc. in Data Science and Machine Learning — *Aug. 2023 – Dec. 2024*

Wuhan University — **Wuhan, China**
B.Sc. in Geospatial Informatics and Digitalized Technology — *Sep. 2019 – Jul. 2023*
GPA: 87.54/100; **Ranking:** 3/52 (2021-22), 16/52 (2020-21) in the annual assessment

Publications

[ECCV '26] GameWorld: Towards Standardized and Verifiable Evaluation of Multimodal Game Agents
Mingyu Ouyang*, Siyuan Hu*, Kevin Qinghong Lin, Hwee Tou Ng, Mike Zheng Shou
- Standardizes multimodal game-agent evaluation across 34 browser games and 170 tasks with paused sandbox execution, dual agent interfaces, and deterministic state-verifiable scoring. Received the Best Paper Runner-Up Award at the Multimodal Digital Agents Workshop (MDA), ECCV 2026.

[CVPR '26] FocusUI: Efficient UI Grounding via Position-Preserving Visual Token Selection
Mingyu Ouyang, Kevin Qinghong Lin, Mike Zheng Shou, Hwee Tou Ng
- Addresses redundancy in UI perception by selecting instruction-relevant UI tokens while preserving positional continuity, improving grounding accuracy and efficiency under aggressive token reduction; received the CVPR 2026 Compute Transparency Champion award.

[AAAI '26] SlideTailor: Personalized Presentation Slide Generation for Scientific Papers
Wenzheng Zeng*, Mingyu Ouyang*, Langyuan Cui*, Hwee Tou Ng
- Generates user-aligned editable slides from papers using example-based preference conditioning, visual templates, and a chain-of-speech planning mechanism.

[arXiv '24] The Dawn of GUI Agent: A Preliminary Case Study with Claude 3.5 Computer Use
Siyuan Hu, Mingyu Ouyang, Difei Gao, Mike Zheng Shou
- Presents a case study of Claude 3.5 Computer Use on diverse desktop tasks, with an out-of-the-box framework and analysis of planning, action, and critic limitations.

[CVPR '24] AssistGUI: Task-Oriented PC Graphical User Interface Automation
Difei Gao, Lei Ji, Zechen Bai, Mingyu Ouyang, Peiran Li, Dongxing Mao, Qinchen Wu, Weichen Zhang, Peiyi Wang, Xiangwu Guo, Hengxu Wang, Luowei Zhou, Mike Zheng Shou
- Introduces a 100-task Windows GUI benchmark and a GUI agent framework for task decomposition, GUI parsing, action generation, and reflection.

[TIP '24] JPEG Quantized Coefficient Recovery via DCT Domain Spatial-Frequential Transformer
Mingyu Ouyang, Zhenzhong Chen
- Introduces DCTransformer to recover JPEG-quantized DCT coefficients with joint spatial-frequency modeling across varied compression quality factors.

Professional Experience

UseIt AI **Singapore**
Co-founder *Mar. 2025 – Oct. 2025*
- Led the GUI agent development and workflow orchestration, focusing on creating, editing, and deploying personal agents.

Sensetime Singapore Research **Singapore**
Algorithm Researcher Intern *Dec. 2023 – Oct. 2024*
- Developed efficient low-level deep learning models for RAW image processing, including noise synthesis and learned image signal processing for built-in camera and smartphone modules.

Tencent **Beijing, China**
Machine Learning Engineer Intern *Mar. 2022 – Oct. 2022*
- Implemented GPS data processing using Hadoop and Spark. Developed and maintained an internal service that automatically executes daily tasks, handling over **10 billion callbacks per hour**.

Springturn Education Technology **Zhuhai, China**
Co-founder, Software Developer *Nov. 2020 – Sep. 2021*
- Co-founded the company and secured angel investment exceeding **¥1 million** in the second year of business.

Research Experience

ShowLab, NUS *Aug. 2023 – Present*
- Research focus: MLLM-based agents for GUI, computer-use, and game environments; agent-environment interaction.

IIP Lab, Wuhan University *Oct. 2022 – Jul. 2023*
- Research focus: Low-level vision tasks, including image super-resolution, denoising, JPEG artifact removal.

State Key Lab of Info. Engineering in Surveying, Mapping and Remote Sensing *Sep. 2021 – Jun. 2022*
- Research focus: 3D scene perception, fusing point clouds and BEV images for road-scene understanding.

Open-Source Projects

[GitHub 1.9k stars] Computer-Use OOTB: showlab/computer_use_ootb
- Core developer and maintainer. A desktop GUI agent framework for Windows/macOS with API-based or self-hosted model and local execution.

[GitHub 1.8k stars] ShowUI: showlab/showui
- Maintainer. A training and inference codebase for a lightweight (2B) VLA model for GUI agents.

Miscellaneous

Wuhan University Football Team
- Served as captain for three years and led the team to win the championship at the 16th Hubei Province Games (2022).

Technical Skills

- **Programming Languages:** Python (advanced), Java (Hadoop/Spark), TeX, HTML/CSS (Vibe).
- **AI/ML Frameworks:** PyTorch, Transformers, vLLM, OpenAI-compatible APIs.
- **Computer-Use Agent Infrastructure:** On-device control (Windows, MacOS), cloud services (AWS, Azure), and agent framework orchestration.
- **Agentic Coding Tools:** Cursor, Claude Code, Codex (approx. 3B tokens/month).